

Yash Gulati

ygulati2@illinois.edu | +1 (925) 542-6205 | www.yashgulati.dev | LinkedIn: iamyashgulati

Education

University of Illinois at Urbana-Champaign

*Bachelors of Science in Computer Engineering | **Minor** in Statistics*

Tech. GPA: 4.0 | GPA: 3.96

Expected Graduation: May 2029

- **Honors:** James Scholar, Deans List (FA25, SP26), Tau Beta Pi National Engineering Honor Society Invitee.
- **Relevant Coursework:** Object-Oriented Programming with Java (Honors), Object-Oriented Programming with C++ (Honors), Electronics & Circuits I, Digital Logic & Computer Systems I, Discrete Structures, Data Structures & Algorithms (Honors), Autonomous Robotics Project, Linear Algebra.

Work Experience

Illini Solar Car Team

Strategy & Telemetry Team Member

Aug 2025 - Dec 2025

Champaign, IL

- Analyzed 240,000+ rows of battery telemetry (voltage, power, current) using Python (Pandas, matplotlib) and SQL to plot power consumption and battery charge, identify anomalies within battery performance, and filter outliers.
- Time-aligned telemetry streams from sensors with differing sampling rates to enable synchronized analysis across MPPT data.

Fiverr.com

Freelance Web Developer

Oct 2021 - Aug 2024

Jakarta, Indonesia

- Delivered 50+ orders building & editing responsive web applications with HTML/CSS and React.js, creating mobile-responsive designs with Figma and full-stack features – Firebase authentication and realtime databases – with a 4.9/5 rating.
- Shipped & managed 7 production-grade landing-page, single-page, and multi-page websites for founders and independent artists.
- Owned end-to-end deployment pipeline – Figma for design, Git for version control and Vercel for hosting with custom domains.

Projects

ArXiv Research AI - Retrieval-Augmented Generation

Python | NumPy | LangChain | Scikit-learn | HuggingFace | Gradio | ChromaDB | Gemini API

May 2026 - Present

- Built a RAG pipeline ingesting over 3.1 million arXiv papers via Kaggle, vector embedding each paper with metadata into a ChromaDB database; calling Gemini API to respond with citation-grounded answers based on the top 4 relevant papers.
- Implemented two-stage retrieval – first a semantic search over abstract embeddings to identify relevant papers, then full-text PDF-to-markdown parsing using PyMuPDF and PyMuPDF4llm at query to give Gemini full access to text from research papers.
- Engineering a citation-grounding system prompt to enforce claim-level attribution and explicit refusal on insufficient context to reduce hallucinations and biases – as measured by RAGAS and BEIR scientific benchmarks for baseline scores.

Technical Skills

- **Languages:** Python, Java, C++, C, Javascript (React.js), SQL, HTML, CSS, Assembly.
- **ML / AI Tools:** PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, LangChain, Hugging Face, LLM APIs (ChatGPT, Gemini, Claude, Ollama), RAG.
- **Tools:** Git, AWS S3, MongoDB, ChromaDB, Gradio, RAGAS, Claude Code, Codex.